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SUMMARY

Product manager for grid-scale BESS with a utility and construction background, eight years at LG Energy Solution moving from systems engineering to product ownership. Originated the 2-hour containerized product that became LG's flagship for AI data center power, with multi-GWh bookings. Works across power systems engineering, techno-economic modeling, and Python tooling to turn site-level constraints into deployable hardware and pricing.

CORE COMPETENCIES

Power Systems & Hardware: Grid-scale and containerized BESS, PCS selection and configuration, UL 9540, black start and islanding, voltage fault ride-through, non-firm interconnection and curtailment, demand response, central plant and data center cooling systems

Modeling & Analysis: Cell-to-system cost modeling, EPC and logistics cost modeling, battery degradation modeling (calendar and cycle aging), system sizing and augmentation planning, Python, Excel

Commercial & Delivery: Product roadmap and business case, pricing strategy, RFP/RFI technical response, vendor and contract manufacturer selection, change order negotiation, utility incentive structuring

EXPERIENCE

LG Energy Solution

Feb 2018 – Present

Senior Product Manager, Grid-Scale BESS

Feb 2021 – Present

- **Roadmap Ownership & Commercial Impact:** Expanded the product portfolio from 4-hour to 2-hour duration with an AC-integrated containerized BESS to capture emerging AI data center demand; secured executive and commercial buy-in to establish the platform as LG's flagship offering for the segment, driving multi-GWh forward bookings for 2027 delivery.
- **Techno-Economics & Pricing Strategy:** Built bottom-up cell-to-system cost, EPC installation, and logistics models to set competitive system pricing, evaluate PCS configurations, and quantify the margin, energy density, and transport advantages of the 2-hour architecture.
- **Technical Translation & System Requirements:** Translated hyperscale and AI data center power constraints into core BESS requirements: authored specifications for compute step-load transient smoothing, voltage fault ride-through with active recovery, gas turbine pairing, black start and islanding, and capacity management for non-firm interconnection with curtailment.
- **Cross-Functional Delivery:** Led PCS vendor selection and owned the UL 9540 listing path; served as the product interface between engineering, sales, and project execution, aligning volume commitments, cost-down trajectory, development and certification schedules, technical documentation, and commissioning and service procedures.

Senior Systems Engineer, Grid-Scale BESS (LG Chem)

Feb 2018 – Feb 2021

- Led product engineering for LG's first prefabricated containerized BESS (2.86 MWh/unit); partnered with contract manufacturer Jabil from architecture through functional prototype, managing cost, safety, and size/weight tradeoffs to replace bespoke field assembly and deliver nearly 1 GWh across CAISO and ERCOT.
- Developed Python and spreadsheet degradation models simulating calendar and cycle aging, C-rate, temperature, and depth of discharge; optimized battery-to-inverter ratios and augmentation schedules to minimize lifetime cost while satisfying developer warranty terms.
- Directed technical sales engineering for integrators, EPCs, and IPP/developers; led requirements capture, system sizing, and RFP/RFI technical defense, translating site duty-cycle constraints into pre-construction proposals.

Comfort Energy

Jul 2016 – Oct 2017

Design-Build Engineering Manager, HVAC

- Managed end-to-end commercial HVAC delivery on active construction sites: scope definition, equipment selection, and mechanical and electrical commissioning and startup, coordinating with general contractors, architects, and developers.
- Owned estimating, procurement, and field operations; produced mechanical bids, directed trade installer crews, and negotiated in-flight change orders to protect margin and schedule.

Pacific Gas and Electric Company (PG&E)

Dec 2013 – Jul 2016

Energy Solutions Engineer

- Delivered 2.5 MW of peak demand reduction and 7 GWh of cumulative energy savings across commercial and industrial portfolios through customer-facing technical audits, incentive structuring, and capital project evaluation.

- Engineered efficiency and demand-response projects for data centers and critical facilities: developed technical scopes and savings models for data center cooling retrofits (hot/cold aisle containment, economizers, VFDs), central plant systems (chillers, boilers, AHUs), and behind-the-meter thermal and battery peak-shifting assets.

SELECTED TECHNICAL PROJECT

Institutional Equity & Macroeconomic Research Engine | Python, Gemini API, PyMuPDF, SQLite, Discord API, Railway
Automated synthesis and cited RAG bot converting institutional research into a daily brief and Q&A for 100+ paid subscribers.

- **Multi-Source Ingestion & Schema Extraction:** Engineered an automated pipeline ingesting 100+ equity and macroeconomic research PDFs daily (16,000+ total); parsed unstructured multi-asset commentary into structured JSON schemas with verbatim anchor quotes, programmatically verifying every extracted metric against source text before database commit.
- **Morning Pulse Synthesis:** Built the automated "Morning Pulse" briefing, published every trading day (110+ issues) with a lead theme, desk-call trade board, and economic and earnings calendar drawn from live options chains and agency APIs; a redesigned synthesis pipeline in shadow pilot raised extraction fidelity against source PDFs from 33–60% to 87–100% and cut grouping fragmentation from 30–60% to 9–16%.
- **Cited RAG Agent:** Built an interactive Q&A agent executing cited answers over the research corpus (1,470+ production queries); regex-based tool router restricts model actions per query, and a payload provenance guard blocks any figure absent from tool output.

EDUCATION & CREDENTIALS

MBA (STEM-designated), Product Management/Strategy, UC Davis Graduate School of Management, 2022

BS Mechanical Engineering, UC Davis, 2013, Dean's Honor List

California Licensed Professional Engineer #M38303, Thermal and Fluid Systems (NCEES), 2016